

Experiment 3.

Characterization and Extraction of Various Parameters of uA441 IC

Aim

To characterize and extract the operational amplifier (OP-AMP) IC-uA741 parameters and compare them with the manufacturer's datasheet values.

Theoretical Background

While an ideal op-amp draws no current and is independent of temperature, a real op-amp exhibits non-ideal characteristics due to transistor mismatches and temperature shifts. The key parameters analyzed in this experiment include:

Input Bias Current (IB): The average of the currents entering the inverting (IB1) and non-inverting (IB2) terminals.

Input Offset Current (IOS): The algebraic difference between the two input bias currents.

Input Offset Voltage (V_IOS): The differential input voltage required to nullify the output voltage.

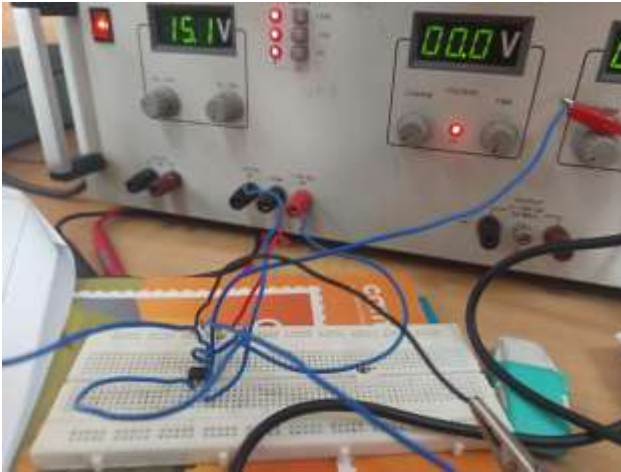
Slew Rate (SR): The maximum rate of change of output voltage, typically expressed in $V/\mu s$.

Observation Table

The following table compares the theoretical datasheet values provided in the lab manual with the measured data from the laboratory session.

SR NO.	PARAMETER	THEORETICAL VALUE	PRACTICAL VALUE
1	INPUT BIAS CURRENT	80nA	18.1818nA
2	INPUT OFFSET CURRENT	20nA	36.3636nA
3	INPUT OFFSET VOLTAGE	2.0V	0.269mV
4	OUTPUT OFFSET VOLTAGE		-0.267V
5	SLEW RATE	0.5V/ μs	0.471V/ μs

Hardware Setup:



Results and Inference

Analysis of Results

Bias and Offset Currents: The measured Input Bias Current 18.1818nA was significantly lower than the typical datasheet value of 80nA , indicating high-quality input transistors in the specific IC tested. However, the Input Offset Current 36.36nA exceeded the typical 20nA value, suggesting a slight mismatch in the internal differential pair.

Offset Voltages: The measured Input Offset Voltage 0.269mV is well within the maximum limit of 6mV .

Slew Rate: The measured Slew Rate of 0.471 V/us is very close to the theoretical value of 0.5 V/us , confirming the IC's performance limits for high-frequency AC applications.

Inference

The characterization of the μA741 IC shows that while the measured parameters generally align with the expected ranges, the discrepancies in offset current highlight the practical limitations of real-world components compared to ideal models. The low slew rate confirms that the μA741 is best suited for low-frequency applications.